

ActInf Textbook Group ~ Cohort 6 ~ Session 14 (Chapter 6, Part 2) ~ 7/22/24

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 14 (Chapter 6, Part 2) ~ 7/22/2024 | 1:04:58

all right welcome back slw welcome back
to cohort
6 couple quick updates as we
mentioned with octopus and others there
are math discussion sessions they're
Tuesdays at 12
UTC octopus session videos has the links
to the
videos so feel free um these should
be awesome freewheeling and super
epistemic
sessions and as with the material on the
book itself all kinds of questions from
the most background and basic and
confirmatory for one on through the most
speculative and and farsighted for for
others it's always a contribution to to
write it down and as the math comes to
play and The Meta math and the math art
it's going to be a great journey
together um today we're in our
second pass on chapter six which is the
recipe for active inference modeling and
also a new or different feature in this
interval of activity compared to
previous weeks is the uh third week
Rhythm so in the
uh third week for cohort 6 going through
the second half of the book will really
focus on applying active inference
Andrew has written a tutorial utilizing
PDP with a focus on agent based modeling
for the social sciences so that should
be pretty cool and additionally as uh RX
and fur development continues we'll be

learning more and more about that um in cohort 7 we haven't quite determined what the third week activity will be in that uh slot so we'll find out uh discuss later today with that cohort and and see what's useful in the third week for now though let's uh jump over to chapter

six we can start with

any section of six any idea or page or figure in chapter

six that anybody wants to bring

up we could look at any of the prior

questions and also as chapter six is

about applying and designing active

inference models it it can be any

question that someone has so just raise

your hand or write it in the chat or

unmute and go for it

okay I'll just give one brief catch up

to get us to six to to get us uh there

in just a minute and then again just

type a question or raise your hand so

part one of the book they outline as

kind of the epistemic and the background

and learning part two is heading kind of

home stretch heading downhill heading

back home in into the applying chapter 1

and chapter 10 are like book ends

they're very similar chapters chapter 10

is a bit more broad ranging and longer

whereas chapter one introduces the ideas

whereas they're similar because they

provide

perspectives in chapter one this concept

of the low road and the high road is

introduced and that's Revisited in more

detail in chapters two and three low

road is like the how and the what it's
the actual architectures that are built
it's the way that base theorem applies
to perception and cognition and action
in a unified cognitive model and then
the high road comes from the free energy
principle physics of cognitive systems
basian mechanics the imperative to
persist to remeasure and so
on chapter 2 and three get us to active
inference and chapter four is where we
actually see the active inference
generative model itself which is what's
going to be given the recipe for in
6 chapter V goes through the generative
model describes some of its analytical
properties and introduces us to the idea
that there's a discrete time and a
continuous time form chapter five
message passing in neurobiology gives a
quick review of one of the systems of
interest that has received the most
study to date in terms of active
inference modeling which is like Maman
and human
neuroanatomy that's part one of the book
part two of the book goes into applying
active inference chapter six is very
provocative and fun and it's a recipe
for Designing active inference models
and as we'll I hope unpack a little bit
today it doesn't require some advanced
level of programming or anything like
that to to get going and actually with
augmented coding with templates and some
other tools there's a lot that can be
done just through conversation with the
domain expertise or familiarity and the

active inference

ontology chapter seven and eight then
are going to focus on some of the
features and examples of discrete time
continuous time models chapter nine
introduces data into the picture how do
empirical data relate to generative
models and then again chapter 10 is a
summary so that's the
textbook it's it's a long pull and then
when we kind of pull back to this level
of course graining it looks really short
too

so what's a section or a question on six
that that anybody would like to go to
okay we'll just go to the questions that
have been sorted by the thumbs
up you can click on the the symbol to
follow wherever uh a person is in Koda
and I'm putting it in the chat as well
so if you want to you can hit thumbs up
for the questions that are interesting
or again just like type it or raise your
hand

okay first

question what are the four steps in the
recipe to construct an active inference
model does anyone want to give a thought
on this

how are these steps similar or different
to any other kind of systems modeling or
analysis technique that people have seen
e e

okay uh John

okay let's look at figure

6.1 John go for it if you want to yeah

yeah sorry I had to set up properly

there um I'm I'm dealing with something

at work right now so I'm gonna be like
in and out on this um but a a question
I've kind of had come up and maybe this
is just from not seeing too many um
active inference agents implemented yet
is sort of how how do you fit expected
free energy in
with um free energy
minimization like for the current state
um versus looking into the future um
it's not something I saw detailed a ton
in chapter six
um but yeah and I guess answering the
question you posed um I think the like
looking at our we don't have you know
reward-based policies but we just focus
on minimization of free energy expected
free
energy um but yeah
like it's a little unclear to me how
those two
interplay um what what their interplay
is in
implementation great great question yeah
it's gonna it's gonna touch on a lot so
if if anybody wants to like add in or or
or going deeper into it some of this
please just write it or raise your hand
um otherwise let's get there so the
question is how do we think about free
energy minimization so the realtime free
energy minimization with equation 2.5
and the prospective expected free energy
minimization in equation 2.6 so how do
those pieces come together in an active
inference generative model okay let's
start with figure
6.1 this is called the particular

partition it's a pun particular means like that specific one also particular means related to a particle and the agent or the cognitive particle is consisting of the internal States and the blanket States so everything to the exclusion of the external States is called the particle blanket plus internal equals particle so it's like the brain and the skin so to speak that's figure separated from ground that's the agent separated from the niche that's called the particular partition and the free energy principle makes this kind of axiomatic starting point that we're going to partition systems according to a blank and the the two directions there so this is kind of the the partition on what we're looking at okay

now how is free energy minimized in in real time and where does planning and prospection come into this we have some external State let's say it's the actual temperature of the room sensory States coming in that's like thermometer readings internal States like internal representations of temperature and then action states which be like to modify the air conditioner that's kind of the setup of the model

David uh yeah I was just wondering how uh discret time would play in the um exchange between the different states in terms of being able to model fit for example yeah great question um the way it comes into play in discret time we think about each of these nodes having

like a number and then there's a discrete tip of the model and like all the numbers update in a tick in continuous time these four states are thought of as being like four simultaneous flows so it's like the difference between like four discretely updating synchronized changes and like four analog control knobs that are kind of always giving a value and then they can change independently okay so now let's look at the recipe then we'll get to equation 2.5 and 2.6 in the recipe first we identify the system that we're modeling this is the cognitive particle that's that's or it's really the cognitive particle plus Niche it's the total setting and then question three is basically how are we going to set up the particle in terms of the blanket and the internal States and question four is how are we going to set up the generative process in terms of the niche so these four questions basically just get us to the point of specifying the agents including blanket and the niche okay so how does framing things in that way relate to the real time inference and uh to prospection part of the answer is that it depends a lot on the system that we're discussing so it's kind of sometimes frustrating the general in general but it's because

it depends so much on the system that we're modeling so here we see a slightly different representation of the particular partition also of course slightly different notation Etc and this is like a Continuum from simpler systems on through systems that have uh introspection or it can have planning but the overall scheme is is very similar okay so John asked where does expected free energy fit in let's look at the equations for it and then we can look at the pmdp documentation to see what it looks like

equation 2.5 is the variational free energy variational free energy it's a functional so it's a function that takes in a function and what it takes in is q a belief distribution and why a new observation data coming in so this is like the agent's belief about temperature distribution that could be a super tight distribution very loose distribution and then Y is a data point and here are three equivalent ways to frame this or to decompose the same quantity the

top uh line is using a physics based ontology of energy minus entropy the middle is using a statistical ontology of complexity minus accuracy which is to say we want to reward models for fitting well however we want to penalize them for having more parameters that's also known as the key information Criterion or the Baye information Criterion basically in that

adding more parameters will always fit the data better so the imperative can't Simply Be Fit the data better otherwise you take that to the end of the road and you just have this massively overfit model so the question is how do you penalize a model for its growth while rewarding it for fitting well and that's the kind of accuracy minus complexity uh trade-off Frontier and then there's a Divergence in evidence representation these are not look ahads this just relates to beliefs at a time point and an incoming data point that's because variational free energy is attractive optimizable bound on surprisal surprisal itself is often difficult to compute because it it can be a large computation however variational free energy is this tractable bound this is not introduced or invented by active inference variational autoencoders any kind of variational inference uses the exact same approach it's known as the evidence lower bound or the variational lower bound so if you can't necessarily fit the likelihood directly to maximize evidence which is minimizing surprise there's this tractable bound so this is not something that that is being introduced by the active inference literature but rather variational methods variational approximations are being applied to the whole iguana to a unified model of perception cognition and

action but equation 2.5 doesn't have any
Planet it's just about beliefs at a
given moment and then what is the sort
of Base optimal move given this new
observation David go for it yeah I was
just wondering if the uh part of the
entropy formula there was related to uh
Heisenberg h
does anyone have a thought on
that you mean
the I'm I'm wondering I don't understand
the question
sorry yeah did you mean like Heisenberg
uncertainty or what yeah the uh the
uncertainty principles so in terms of the
uh functions itself related to Shannon
entropies
yeah let's look into it
more there there may be some connections
yeah it just it does deal with um fast
Warrior transform so I was just thinking
in terms of um major
modeling cool yeah in interesting
possibility the 2022 textbook doesn't go
into the
quantum uh
interpretations however maybe maybe we
can look more into it okay so this is
great for
modeling uh bound approximation yeah go
for it David oh sorry oh sorry I Wasing
bable approximation for the situation of
beliefs and a new data point coming in
so that can apply to trivially simple
systems like that have no reactivity or
sentience at all or kind of classical or
conservative active systems like um you
know a given data point comes in it just

it just makes a linear adjustment in a belief it also describes that real time flow on more complex systems however it doesn't invoke any future time points this is just kind of like the slice of the now so when we were interested in modeling agents with planning that is when we move into the expected free energy so expected free energy is a cousin of variational free energy it's G it's also functional um instead of f it's like the one after FG its um argument that it takes is Π Π is the policy prior so it's a vector of all of the possible policies affordances are actions that can be taken in a moment up down left right policies are sequences of affordances for a given time Horizon so for a Time Horizon of one the policy Vector is the same as the affordances for a Time Horizon of two the policy Vector is of the length affordances squared and so on so it's like up down left right are the affordances those are the opportunities for action in The Niche policies could be just one one time step if there was no planning just action selection or policies could be longer like up up down up up right um so that's about the defined time Horizon for the model which we'll look at in pmdp um after this G takes in the policy prior and it's going to update it according to how well these policies score on expected free

energy what gives a good score for expected free energy here we see several decompositions there's kind of like three and a half decompositions because the the second and third line are really similar the first decomposition is in terms of epistemic value or expected Information Gain and pragmatic value or utility so a lot more to say on this just just going quickly but pragmatic value in terms of utility is the expected alignment between future observations why Tilda and preferences their preferred observations whereas epistemic value is about the KL Divergence between double line beliefs about hidden States Through Time Q on X Tilda conditioned upon policy either way and observations so if the observations coming in in are not expected to update beliefs these two are the same and the KL Divergence is zero Information Gain is zero if expected observations coming in are expected to update beliefs like I want to know what's on page 12 of the book what about the policy to look on page 12 of the book I expect those observations to give me information even if I don't know what direction the information is going to be updated in and that is assigned a high epistemic value this is a point of differentiation with reinforcement or utility based models which is those kinds of models only have pragmatic

value or utility so they need to discover these kinds of tricks and methods to give a utility to Information Gain and there's all kinds of tricks you can have heuristics like mix it up every 500 times steps or could say give this price on information put this price on basic research whereas here they're kept separate um and there's some other representations but what's important here is whereas

2.5 was about our beliefs and then the data point coming in in a moment expected free energy is arbitrarily forward-looking for a given time Horizon

so a good example of where this comes into play is in this um active inference epistemic chaining demo I'll put it in the in the chat as well and probably um linked elsewhere

um

okay well since we're in six and actually like looking at how these models are coming into play let's look at this script but again just raise your hand or or type it if you have any thoughts and also note this is the PDP python kind of uh script so RX and fur is quite different in its style but this is what PDP looks like okay

first like many other Python scripts different packages are imported those can be

core packages or other broad packages like numpy and it also includes specific functionality from elsewhere in the pmt package um so we can kind of read this in light of the of the recipe so just to remind yourself system of Interest like situation of Interest what does the agent look like in terms of of its perception cognition internal States and action and then what does the environment look like so it's not going to be in that order in the script but we'll see that all those boxes will get ticked okay first they set up the map so this is like declaring the generative process and it's this grid world where there are certain cells that have Rewards or it could be just called food and then there's some that have cues which are like signals here they use M plot lib to visualize the grid so in in this world there's two possible places where there could be food or not so it's kind of like a t-maze however the reason why it's called epistemic chaining is that there's different places where information can be found and that information reduces uncertainty about where the food is so that's the generative process they you should call grid world now we get to generative model okay we're in discrete time here um pmdp is is uh only in discret time here's figure 7.1 so a little looking

ahead to discrete time model this is the core sense making Motif this is what makes something partially observable is we have observations coming in at each time point the observations are mapped through a to Hidden State s b describes how hidden States transition through time at each time step and D is the prior just to kick off the chain so in the neuroimaging setting like SPM observations are the the noisy sensors and then s are these latent activity patterns in the brain that's partially observable inference that's like a Kalman filter active inference is going to drop in on top and add this upstairs bit but the downstairs is exactly the saying that's the sense making motif of observations kind of getting mapped into latent states where the action comes into play is when policies intervene in how things change we have like observations like thermometer readings coming in at each time step and they're getting mapped to uh beliefs about temperature in the room and then B is a matrix that is going to apply to the hidden State and fold it into the next time step so then it B has these slices it's kind of like index cards and policy selection by the agent is like picking which index card is going to be applied to unroll into the next state so one of the index cards might be keeping something like just in the location setting keep something where it is no move then that would keep s the same

whereas there might be turn on the air conditioner move left and then those would be represented in the structure of slices of B

this works in terms of uh weights and values for u_m each portion because you had mentioned u_m with active in being applied at G as a bit would that be considered u_m adding any data weight to u_h to the formulas

okay yeah good good question so again downstairs is our like perception and sense making here upstairs is our policy selection applying variational inference evidence lowerbound variational free energy to the downstairs sensemaking part is like a variational auto encoder the move in active inference is to say agents are described by The Joint distribution of their belief about perception cognition and action and then we're going to take a variational free energy on all of that so that's the unified modeling angle that is u_m what unifies like the perception inbound signal processing problem with the outbound control theory problem so then the question is like well okay how how do this is a structural model how do you get at the weights let's see how they do it in the demo okay so here they're setting up the size of Frank here they're setting up the size and the dimensionality of observations which are called

OBS and hidden State factors which are called numb States so an important thing

to to note is that

um this is like the schema or or the
sort of template

Motif however let's just say that

each grid cell uh each cell in the grid
had like a temperature a smell and a
light

variable there's modular choice in how
that is

represented you have three observations
coming in and then you could imagine
like those all project onto one hidden
State

type or you could set it up so that like
there was a hidden state for each of
those sensory

modalities and then those were or
weren't integrated so this is like map
territory this is why Toby St Clair SMI
calls this

work compositional cognitive

cartography compositional because we're
playing with Legos in category Theory
and we're composing

models cognitive because we're talking
about cognitive interfaces and things
perception cognition action and then
cartography because it's a science or
study of map making so this isn't to say
that like if you broke open an animal
you'd find this base graph it's to say
that this is a map like a subway map
that we making that has all of these
really tractable properties so there's a
lot of

places for modular degrees of freedom
and there isn't necessarily like a
correct map of a cognitive system

anymore than there would be a correct map of of a city here they're setting up the variables just in terms of their length without going too much into RX and for at this moment one of the major challenges SL frustrations with p DP and with modeling is the dimensionality and making sure all the dimensionalities work out otherwise you get like just a bunch of matrices that get multiplied and the shapes don't work and it just kind of like gives bugs

okay next they're going to State a a Maps between observations and hidden States so this is the downstairs sense making part the a matrix they make sure that its shape in terms of rows and columns is basically related to on one dimension the number of hidden States and on the other dimension the number of OB the type of observations so you could have five mapping on to five or you could have one mapping on to five or you could have five you know any any combination then they fill out um all Zer just to get it uh

going here there's three modalities on a so a is kind of a is a tensor each slice goes from observations to Hidden States within like one frame so you can think of it like three index cards or like three sheets in an Excel file or a tensor that has three uh stacks

a a Zer is location A1 is q1 A2 is Q2 this is zero

indexing so it's kind of like the first element is the zero just like birthday and so the numbers are being filled in and then there's this um third third modality A_3 which is the observation of reward or not so in how these matrices are built is where a lot of the nitty-gritty happens in in a okay now they move to specifying b b is the transition model so this is how do those hidden States change through time again there's the work of making sure that the sizes and the shapes of all the variables line up and then all of the work of like what does Action do all the action representation is loaded into B and it's like when you see the model it's like there's actually not many places that that uh again in this simple form there's not many places where learning can go but it's important to see that a is action independent a is the mapping from observations to Hidden states that can be a recognition Direction recognizing or it can be a generating Direction like synthetic data all of the efficacy of P_i all of the efficacy and the differences that make a difference with policy selection influence how things change through b so here's where we see the actions having an impact in terms of different be slices that can get pulled out in the b_0 which is um location just like a z was location we we have efficacy of actions

actions should change our location
however there's no action that the mouse
can take to change the q's and so here
the B um variables in that first first
and second modality are called
uncontrollable because there's no policy
that makes a difference for those
modalities preferences are
specified we can talk about preferences
but they play the role in contributing
to pragmatic value calculations if you
don't have any uh preference or you have
a flat preference like outcomes are
equally desirable
then epistemic value or learning or
novelty comes to dominate The behavior
whereas conversely if you have hyper
precise priors there's no epistemic
opportunities and strong preferences you
get the special case of utility
maximization then the prior over D is
specified so that was a b c d and e it
was just framed in terms of the actions
basically and any thoughts on
this okay just like in the recipe they
then turn to the the rules of the
game drawing inspiration from machine
learning and reinforcement learning all
this they use an EnV for environment
class which steps in discrete time one
by one so we defined the agent in terms
of ABC D and then they write this grid
World environment
example which is kind of like um we we
can look in more detail but this this
actually is the consequences of action
whereas before when we were defining the
agent this is its beliefs about action

what that will do and then here are the actual consequences of choosing actions so this begets a question like well H how does the agent come to know the structure the consequences of its action that's like not addressed in this notebook but like it's like playing a video game and then like you're um you were used to having up on the joystick move you forward but then now you're in a inverted mode where it moves you backwards so just because the agent is specified to have a given beliefs like oh if I hit this button the room will get colder but that's not necessarily A veritical belief but that's what the modeler is if if they're interested in this question of how do Agents come to learn the consequences of their action maybe you'd set up a different model than I'm studying agents who already know the consequences of their auction just like what are you trying to ask who's the model 4 all those early questions of chapter 6 okay all right now we get to the active inference part initialize an agent and an instance of grid world again this is like chapter six we defined the the situation of Interest then made the agent ABC D alphabet and then defined the environment the generative process grid World M my agent is specified and here the arguments it's taking looks a little bit like redundance but that gives you like a degree of freedom in specifying the agent so you could say I want a and then

I want to customize it and so this is just a wrapper method that bundles together a b c and d and policy length so that's where the expected free energy is going to come into play Because at each time step it's going to compute uh variational free energy but the policy length is used for the expected free energy so in this situation you have five affordances up down left right and stay and there's four time four four time steps is the time Horizon of consideration so that means the policy Vector is going to be $5 * 5 * 5 * 5$ and it will sum to one here's another point a difference to to emphasize with reinforcement learning so that's called this the policy space Π is the space of all policies that can be selected from in reinforcement learning or reward learning there's another function proposed called the utility or the reward and it is going to give a value to each of those points in policy space so each policy is going to be given a value and then that is going to rank the policies and then you could choose the most rewarding most valuable policy or you could you could probabilistically select from policies according to their expected value in active inference there's no proposal of this auxiliary reward function we have our policy prior which is like our

habit and then whether we apply G or not
which updates to the policy posterior
that policy variable is directly sampled
from so policies are selected based upon
their likelihood either they're the
likeliest thing that would happen so we
just pull from the top of the list
that's kind of like the most rewarding
policy but rather than moving through
this intermediary auxiliary function
we're picking directly either the best
policy the likeliest policy or
probabilistically choosing from the list
that's the whole physics of the decision
making angle is there's no secondary
reward assessment it's just updating the
policy variable and then choosing from
it okay spin up the
environment and reset it just clean up
the
environment now here's the actual active
inference Loop so it's cool because
python has this semi pseudo code
feel history of locations is just
creating a trace or a log
variable and the number of time steps is
specified it's 10 time steps so it's
going to run for 10 steps but the
agent's window is
four so while we're within the range of
steps this is like where we get to kind
of the UDA like observe Orient decide
act because we're actually going to see
procedurally what happens within the
active inference Loop first
 Q of s that
was Q of s is Q of x q of hidden
States so here it's it's taking in

observations and it's inferring hidden

States that's Q of s

then there's policy inference so this is

sense making that's the downstairs and

then there's control that's the upstairs

and then after updating the policy prior

into the policy

posterior the actions are

selected the

movements the The Chosen action is is

kind of locked in so even though there

was a probability distribution over the

five options for movement eventually one

of them selected and this just clarifies

yeah which integer was it was it 0 1 2 3

or

four and then the implications of that

action

occur this prints a line to the terminal

just to log what

happened then the environment takes a

step predicated upon the choice of the

action of the agent and then there's

just an updating of some of the

observations and the history gets

appended

to so this is what it looks

like visualization

method so here the the agent starts here

it comes up uh oh starts in the top

left it steps here to determine which of

L 's have the Q that's

informative it navigates to there and

then told it that the bottom spot was

the one that would have the preferred

observation

David So in terms of um modeling the map

itself and updating the agent priors for

the expectations um is there like a
library call for a different environment
or is it going to be um like a different
kind of data store for
example yeah there's a lot of ways to do
this like impdp RX Ander Etc um one
really useful articulation is right here
so let's just say that the final um
agent method is going to take on a b CDE
e one convenient approach is have like
um a bunch of a like a bunch of a
Alternatives that you generate you know
a um you
know call yeah a noisy and then and then
you make an agent Constructor and let's
say you have 10 options for a matrices
and 10 options for B matrices then you
make this a wrapper method that says
okay now spin me up all 100 agents all
10 x10
combinations of agents who have a matrix
1 through 10 and B Matrix one through 10
and then using these other kind of
methods like visualization and parameter
sweeps then you use those sorts of
methods to sweep across
combinations more modifications here
they move the two cues to hear or the
two rewards to hear um
so it's it's it's an interesting
experience but that's what chapter 6 is
like to see ultimately there's more
lines of code in this related to just
the visualization
methods sometimes it's it's
quite Curt and
concise what is actually being
specified so it's not like we're making

Mega sized models or needing to Tinker
with a wrench on a billion
parameters that's one of the
interpretability elements of active
inference
so that was their that that that was the
output of the recipe here's the recipe
to
follow just at a qualitative structural
level that may have a lot of similar
arities with other systems modeling
techniques people have learned about
like from complexity science or systems
Dynamics or other other strategies of
modeling and then we take the situation
of interest and then basically decompose
it according to the particular partition
that's what grounds this all under the
free energy principle and then more
specifically using the active ontology
so that we know that we've gotten like
the a the B the C and the D the
observation and the hidden state with
those in place you just write them in
using language models or code models to
help and
then the other functions are are shared
across
models
um in our last bit what are any like
thoughts or questions on
that e
here there there what they call a recipe
here this is like more like the UDA Loop
this is just getting into the procedure
of what happens at each time step but
also it's like just like if the
procedure for the recipe was cup of

flour a cup of sugar Etc and you had all those ingredients ready you'd be able to make the recipe so the kind of functional food we're making is an agent that's able to do perception cognition action so if we have all these ingredients we have all of the precursor materials to do this observation sampled from the environment move that observation into the hidden

States through the a matrix through variational free energy minimization like what should I believe about temperature given this thermometer observation and my prior equation 2.5 calculate the expected free energy of actions equation

2.6 policy prior reweight the policy prior according to expected free energy epistemic plus pragmatic value get the policy posterior sample from it or just pick the best one or sample probabilistically from it reenter that sampled action into the generative process that's

it and is there a similar recipe book in terms of rxm so

um we uh want to be building them making an RX and F model um so we copied in like the same recipe should apply okay like chapter six is out a generality of like prepare the main course prepare the drinks set the table but then of course it's all situational what's the main

course what's the drink what's the
what's the main table um but uh we have
Google collab notebooks drawing from
kass's code examples at learnable Loop
so if if and I'll put these these links
in the chat too so the here you can um
jump in to a
a coding
notebook and then like get all the way
on through to to seeing it in RX and fur
like here it is with running a
drone um there's some pretty major
structural differences between PDP and
RX and fur in terms of how you set the
table specify the main course Etc
um and if people want to learn more they
could explore that document and then
also
um we're working to have all of the
textbook
examples
um for the
textbook in RX infer so let's see where
where um 1.2 is
okay figure
2.1 so this is not a case of active
inference this is just a passive
inference there's beliefs about frogs
and apples like how prevalent they are
in the world that's the prior belief by
hidden States then there's a likelihood
model which is like the a matrix and
that is a mapping it's a belief about a
mapping about jumping and not jumping
and how they're related to frogs and
apples so it's kind of a 2 by two shown
here with these two bars like apples
jump 1% of the time frogs jump 81% of

time before we saw the observation we believe that 10% of the items in the world were frog we observed a jump and then we basically multiplied this shape by this shape and it turns out that a jump is 81 times more likely to have come from a frog than an apple just 81% versus 1% however we believe that apples were nine times more likely so um something was it was it was 9x down in in in the um odds and then we observed something that by 81 fold made it more likely so it went from 9x down to 9x up so that's the basian update and then this is what it looks like in RX and fur so just really briefly here's installing Julia so that's not related to aroner here's install um just showing okay have we installed Julia here's installing the RX and fur package and other needed tools frog or apple it's a categorical distribution with 0.9 and 0.1 right here 0.9.1 here's the T Matrix that's that's what it's just being called here but the variable names don't matter it's what what they are and then there's this observation that takes in those two variables and and updates based upon observation and then in RX Ander just calls you just call it and you say the data were surely a jump and then you return the posteriors so it's a different

style and it's one that that many of us are still learning um however when we have well documented examples of everything in the textbook in RX and fur it'll bring a lot of continuity between the kind of engineering grade application framework and the learning path so for for anybody who wants to work in that way it it's super helpful um okay I hope these have been some useful pieces for people any last this um I don't want to go over time hear too much or anything but um one thing I have been thinking about is how how you actually distinguish between a continuous versus a discrete space or time um in in that like there are you know these limitations on your inference procedure um all of that calculating your your expected free energy and and making a decision taking an action so you're kind of living in a discrete space that's dependent on your computational model um when you're working in the continuous space um which to me also raises a question on like in referring a policy about your temporal breadth of of of like the now um which is something that we kind of do with adrenaline right so maybe maybe we or at least it feels so um yeah I don't know that that's just another dimension of modeling you're you're right is it it's another dimension or layer and like that's part of the fun and the

open-endedness of having such a
composable framework like temporal
thickness or um anticipation of future
agency over consideration of time
thickness like that's not here but if
you can dream it you can do it so you
could make those kinds of structural
hypotheses explicit so whether it's
making a map of a cognitive system that
does act as if it has these properties
or if you're just on a flight of fancy
or a flight of utility and you're like
could we make a policy variable
controlling this and then you just add
another node in an
edge crank it again so that's that that
is the fun is saying okay
then how how would we do that if there
was an agent that was observing it
adrenaline levels and the adrenaline
actual levels were
modifying temporality this
way maybe that's a map of
nowhere maybe it's a useful map so that
that is the whole
question and and that is why there's
such a space between the
fundamentals like the particular
partition variational free energy
expected free energy like you can't say
too much more generically than those
equations the whole point is those
equations whether in real time or in the
anticipation those equations apply to
whatever graph is
constructed but that graph could be
anything could be um a lattice of just
ones and zeros could be a continuous

automa so it's like it doesn't even have to resemble something agentic and like Maxwell said in Discord several weeks ago but but I but think about all the time he said the focus on agents has reduced the apparent scope of active inference because agent as we know it is not anywhere necessarily here now one interface that's interesting is that like cybernetic perception action interface but just more generally we could talk about like Fabrics of message passing or ecosystems of shared intelligence

so there's kind of the the fundamentals and then it's all about what you build and and how the models really play out in in their environment

so that's kind of

fun and it allows the the scaffolding externalization of cognitive hypotheses and the consideration of alternate models but like the Lego set is not opinionated on

architecture right that makes sense to me yeah thank you for the answer cool yeah it's a good question okay so um if this time works for people then next uh week will return in the third week and that we'll we'll see if these are um in the cohort 7 recorded or not but we'll see um who has a feeling on what these sessions could be about maybe we revisit octopus math group developments maybe we kind of look at recent updates look at philosophy look at other uh areas and then in um the third week for Co

six especially with Andrew um giving the
tutorial at the workshop um and at some
conference and and uh RX and fur and
everything like we'll get down to it
with really applying it including like
everybody can make a copy of the collab
notebook so like everybody can be
running it and it really demystifies a
lot when you see it run in your browser
and you see which variables are defined
how do the variables interact what did I
put into this what do I get out of
it so it's
cool
okay thank you very much see you all
next time thanks have a good one bye